Sahaj Bhadja

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Summary

Software developer with 2+ years designing, developing, and debugging **database-backed applications**, APIs, and engineering tools; shipped Python/Java/JavaScript systems handling 2M+ records and 10K+ daily requests with 99%+ production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and developed **Python FastAPI services and relational schemas** for 500+ users, implementing row-level isolation, indexed queries, and optimized access patterns that reduced application latency from 1.2 seconds to 350 milliseconds.
- Troubleshot **database and application workflows** across 5,000+ monthly jobs, implementing validation, reconciliation, error handling, and failure isolation to achieve 99%+ reliability and accelerate diagnosis of recurring production defects.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **full-stack React and Flask application features** for uploads, processing, history, and reporting, supporting 150+ users and 10,000+ daily requests within an existing Blueprint-based MVC software architecture.
- Debugged **API, file-processing, and model integrations** by adding validation, usage logging, persistent results, fallback handling, and concurrent execution, processing 1,000+ files per batch while reducing turnaround time 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Designed **object-oriented API and data components** using Singleton and Factory patterns, AWS Glue, S3, SQL, and caching, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability.
- Automated **cross-platform builds, deployments, and 20+ tests** with GitHub, Jenkins, PyTest, and Mockito, generating coverage and failure reports while supporting reliable enhancements across Windows, Linux, and macOS applications.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **React, Node.js, and Azure SQL enhancements** across eight microservices for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through REST APIs and WebSockets across locations.
- Implemented **automated testing, monitoring, and staged deployment** with Jest, Selenium, Sentry, Azure Monitor, Docker, and Jenkins CI/CD, cutting release time 50% while improving code quality and production supportability.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Python Flask application APIs** covering bookings, pricing, discounts, payments, and real-time status updates, maintaining sub-200-millisecond responses across 50,000+ records through Firebase synchronization and validated webhook processing.
- Translated **client requirements into software enhancements**, mapping 12+ integration points and implementing cross-platform functionality through Spring Boot, Firebase, unit automation, and version-controlled releases to improve delivery speed 22%.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, Bash, HTML, CSS
Applications: FastAPI, Flask, Spring Boot, Node.js, Express.js, React.js, REST APIs, WebSockets, MVC
Databases: Oracle Database, PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, Firebase, schema design
Engineering: Software architecture, OOP, data structures, algorithms, debugging, testing, performance optimization
Tools: Git, GitHub, Jenkins, GitHub Actions, CI/CD, PyTest, Jest, Docker, AWS, Azure, Linux

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered database application** supporting Oracle and 10+ SQL/NoSQL systems, accelerating query authoring, schema exploration, and table management through a cross-platform Java backend and TypeScript interface with Docker workflows.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed a **containerized Python application API** on AWS ECS/Fargate with CodePipeline, automated validation checks, and CI/CD releases, achieving 87% accuracy and production-grade operational reliability at scale.

Big-Brain Live | *PostgreSQL, GCP, REST APIs, FAISS*
- Built a **cloud database and document application** processing 50,000+ records with incremental indexing, metadata persistence, REST APIs, authentication, semantic retrieval, and performance tuning, reducing data-access latency 40%.